

MyTZStudies.com

Your Free Tanzania Exam Library

Past papers, answer keys, and study resources
for Tanzanian students.

MyTZStudies.com

Past Exam Papers	Worked Answer Keys	Free Access
Choose Your Level	Study Step by Step	Learn at Your Pace

Standards 1-7 | Form 2 | Form 4

Share this with a friend.
Every student deserves free study materials.

MATHEMATICS

STANDARD FIVE

Monthly test - July 2023
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Five
Year	2023
Exam Board / Issuer	Brainmaster Tanzania
Exam	Monthly test - July
Type	Answer key and worked solutions
Questions	47 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S5-2023-Brainmaster-Tanzania-July-Monthly-test.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Mathematics

1. Name the place value of 7 in 70965.

Step 1: First of 5 digits is ten-thousands.

Final Answer: Ten-thousands

2. Write six hundred and six thousand and six in figures.

Step 1: $606000 + 6 = 606006$.

Final Answer: 606006

3. Write 34.654 in words.

Step 1: Read 34 then each decimal digit 6,5,4.

Final Answer: Thirty-four point six five four

4. Rearrange digits 7968 to form the largest number.

Step 1: Arrange digits greatest first: 9,8,7,6.

Final Answer: 9876

5. Name the triangle with labels 20 cm, 20 m, 20 cm.

Step 1: The drawing is a triangle; literal 20 m = 2000 cm exceeds 20 + 20 cm, so lengths cannot form a triangle.

Step 2: If 20 m is a typo for 20 cm, all three sides are equal.

Final Answer: Triangle; intended equilateral triangle if 20 m should be 20 cm.

6. Juma buys 12 plates at 4200 sh each, 11 spoons at 240 sh, 9 bowls at 28040 sh, 4 trays at 8600 sh. Total cost?

Step 1: $12 \times 4200 = 50400$.

Step 2: $11 \times 240 = 2640$.

Step 3: $9 \times 28040 = 252360$.

Step 4: $4 \times 8600 = 34400$.

Step 5: $50400 + 2640 + 252360 + 34400 = 339800$.

Final Answer: Sh 339800

8. Calculate IV/VIII + III/VIII in Arabic fractions.

Step 1: IV=4; III=3; VIII=8.

Step 2: $4 + 3 = 7$; denominator 8.

Final Answer: 7/8

9. Use prime factorization to give all factors of 72.

Step 1: $72 = 2^3 \times 3^2$.

Step 2: Choose powers of 2 from 0 to 3 and powers of 3 from 0 to 2, multiply every combination.

Final Answer: Prime factors $2 \times 2 \times 2 \times 3 \times 3$; all factors 1,2,3,4,6,8,9,12,18,24,36,72.

10. Identify the improper fraction from $11/3, 3 \frac{2}{3}, 5/6, 0.69$.

Step 1: $11 > 3$ so $11/3$ is improper.

Step 2: $3 \frac{2}{3}$ is written as a mixed number.

Final Answer: 11/3

11. List multiples of 6 strictly between 12 and 39.

Step 1: $6 \times 3 = 18; 6 \times 4 = 24; 6 \times 5 = 30; 6 \times 6 = 36$.

Final Answer: 18,24,30,36 (include 12 only if endpoints are intended inclusive).

12. Give first 3 positive multiples of 8.

Step 1: $8 \times 1 = 8$; $8 \times 2 = 16$; $8 \times 3 = 24$.

Final Answer: 8,16,24

13. Continue 29,23,19,[],[].

Step 1: These are primes in descending order.

Step 2: The next lower primes are 17 and 13.

Final Answer: 17,13

14. Give first 2 positive common multiples of 6 and 7.

Step 1: $6 = 2 \times 3$; 7 is prime.

Step 2: $\text{LCM} = 2 \times 3 \times 7 = 42$.

Step 3: Next common multiple $2 \times 42 = 84$.

Final Answer: 42,84

16. Write eight hundred thousand two hundred and eighty-eight in figures.

Step 1: $800000 + 200 + 80 + 8 = 800288$.

Final Answer: 800288

17. How many primes lie from 30 to 40?

Step 1: Reject even numbers, 33 and 39 (divisible by 3), 35 (divisible by 5).

Step 2: 31 and 37 remain.

Final Answer: 2:31 and 37

18. Pulupu shares 45 goats equally among 7 children. How many per child and left?

Step 1: $7 \times 6 = 42$.

Step 2: $45 - 42 = 3$.

Final Answer: 6 goats each; 3 left

19. Rectangle area 180 cm^2 , one side 100 cm , other $X \text{ cm}$. Find X .

Step 1: $\text{Area} = 100 \times X$.

Step 2: $X = 180 / 100 = 1.8$.

Final Answer: 1.8 cm

20. Kiumbo stays 3 days. How many hours?

Step 1: $3 \times 24 = 72$.

Final Answer: 72 hours

21. How many angles in a pentagon?

Step 1: One angle per corner; there are 5 corners.

Final Answer: 5

22. Find sum of primes in 12,13,14,15,16,17,18,19.

Step 1: Primes 13,17,19.

Step 2: $13 + 17 + 19 = 49$.

Final Answer: 49

23. Rectangle 55 m long, 25 m wide. Find distance around it.

Step 1: $\text{Perimeter} = 2(\text{length} + \text{width})$.

Step 2: $55 + 25 = 80$.

Step 3: $2 \times 80 = 160$.

Final Answer: 160 m

24. Dogs bark every 4,6,12 minutes. When next together?

Step 1: $4 = 2^2$; $6 = 2 \times 3$; $12 = 2^2 \times 3$.

Step 2: $\text{LCM} = 2^2 \times 3 = 12$.

Final Answer: 12 minutes

25. Continue seedling heights 11,16,21,26,31 cm. Find sixth.

Step 1: Increase 5 cm; $31 + 5 = 36$.

Final Answer: 36 cm

26. Bells ring every 6,8,12 minutes. When next together?

Step 1: $6=2 \times 3$; $8=2^3$; $12=2^2 \times 3$.

Step 2: $LCM=2^3 \times 3 = 24$.

Final Answer: 24 minutes

27. Find reciprocal of $3 \frac{1}{7}$ (printed as mixed number).

Step 1: $3 \times 7 + 1 = 22$; $3 \frac{1}{7} = \frac{22}{7}$.

Step 2: Swap numerator and denominator.

Final Answer: $\frac{7}{22}$

28. Which is greater, $\frac{3}{6}$ or $\frac{2}{5}$?

Step 1: $\frac{3}{6} = \frac{1}{2} = \frac{5}{10}$; $\frac{2}{5} = \frac{4}{10}$.

Step 2: $5 > 4$.

Final Answer: $\frac{3}{6}$

29. Mkiringa has 897 pupils,45 teachers,50 staff. Total people in Roman numerals?

Step 1: $897 + 45 + 50 = 992$.

Step 2: $900=CM$; $90=XC$; $2=II$.

Final Answer: CMXCII

30. List all positive factors of 24.

Step 1: Factor pairs 1×24 , 2×12 , 3×8 , 4×6 .

Final Answer: 1,2,3,4,6,8,12,24

31. Find the difference between 5 and 8.8.

Step 1: Subtract smaller from larger.

Step 2: $8.8 - 5 = 3.8$.

Final Answer: 3.8

33. Give the value of 9 in seven hundred nine thousand five hundred twenty.

Step 1: Number 709520;9 is thousands.

Step 2: $9 \times 1000 = 9000$.

Final Answer: 9000

34. Find $\frac{5}{10} + \frac{3}{5}$.

Step 1: Common denominator $10: \frac{3}{5} = \frac{6}{10}$.

Step 2: $5 + 6 = 11$.

Step 3: $\frac{11}{10} = 1 \frac{1}{10}$.

Final Answer: $1 \frac{1}{10}$

35. First 3 positive multiples of 13?

Step 1: $13 \times 1 = 13; 13 \times 2 = 26; 13 \times 3 = 39$.

Final Answer: 13,26,39

36. Pili is 8 years older than 10-year-old Jane. Pili age?

Step 1: $10 + 8 = 18$.

Final Answer: 18 years

37. List even multiples of 3 from 3 through 15.

Step 1: Even multiples of 3 are multiples of 6.

Step 2: 6 and 12 are within range.

Final Answer: 6,12

40. Kanana has 9 litres and gives $\frac{1}{9}$ litre per guest. Number of guests?

Step 1: $\text{Guests} = 9 \div (\frac{1}{9})$.

Step 2: $9 \times 9 = 81$.

Final Answer: 81

41(a). Addition table row 200, column 100 is B. Find $B+500$.

Step 1: $B = 200 + 100 = 300$.

Step 2: $300 + 500 = 800$.

Final Answer: 800

41(b). Addition table row 100, column 300 is A. Find A.

Step 1: $100 + 300 = 400$.

Final Answer: 400

43(a). Graph: Mnauke 800, Mery's 650, Nakahako 500, Mkuna 700. Which enrolls most?

Step 1: 800 is greater than 650, 500, 700.

Final Answer: Mnauke

43(b). Graph: 800, 650, 500, 700 pupils; 698 are girls. Find boys.

Step 1: $800 + 650 + 500 + 700 = 2650$.

Step 2: $2650 - 698 = 1952$.

Final Answer: 1952 boys

44. A figure has perimeter 480 cm and one side 150 cm. Find other side.

Step 1: The shape is not named.

Step 2: If it is a rectangle: $480 \div 2 = 240$; $240 - 150 = 90$.

Final Answer: 90 cm if rectangle; not uniquely determined for an unspecified figure.

45. Find a number dividing both 14 and 28.

Step 1: Factors of 14:1,2,7,14; all also divide 28.

Step 2: Greatest is 14.

Final Answer: Common divisors 1,2,7,14; greatest common divisor 14.

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Calculations

7. Kagembe has 846 goats,8076 sheep,405 chickens. Total animals?

Step 1: $846 + 8076 = 8922$.

Step 2: $846 + 8076 + 405 = 9327$.

Step 3: $846+8076+405 = 9327$.

Final Answer: 9327 animals

15. Kalunde and Halima share 600000 shillings equally. Each amount?

Step 1: $6 / 2$: write 3; $3 \times 2 = 6$; $6 - 6 = 0$.

Step 2: $0 / 2$: write 0; $0 \times 2 = 0$; $0 - 0 = 0$.

Step 3: $0 / 2$: write 0; $0 \times 2 = 0$; $0 - 0 = 0$.

Step 4: $0 / 2$: write 0; $0 \times 2 = 0$; $0 - 0 = 0$.

Step 5: $0 / 2$: write 0; $0 \times 2 = 0$; $0 - 0 = 0$.

Step 6: $0 / 2$: write 0; $0 \times 2 = 0$; $0 - 0 = 0$.

Step 7: $2 \times 300000 = 600000$.

Step 8: $600000 / 2 = 300000$.

Final Answer: 300000 shillings

32. 137932 pupils, 5472 guests, 1379 staff attend. Total?

Step 1: $137932 + 5472 = 143404$.

Step 2: $137932 + 5472 + 1379 = 144783$.

Step 3: $137932 + 5472 + 1379 = 144783$.

Final Answer: 144783 people

38. 240 pupils share 12 classrooms equally. Pupils per class?

Step 1: $24 \div 12$: write 2; $2 \times 12 = 24$; $24 - 24 = 0$.

Step 2: $0 \div 12$: write 0; $0 \times 12 = 0$; $0 - 0 = 0$.

Step 3: $12 \times 20 = 240$.

Step 4: $240 \div 12 = 20$.

Final Answer: 20 pupils

39. 15 sweets cost 4500 shillings. Cost of 1?

Step 1: $45 \div 15$: write 3; $3 \times 15 = 45$; $45 - 45 = 0$.

Step 2: $0 \div 15$: write 0; $0 \times 15 = 0$; $0 - 0 = 0$.

Step 3: $0 \div 15$: write 0; $0 \times 15 = 0$; $0 - 0 = 0$.

Step 4: $15 \times 300 = 4500$.

Step 5: $4500 \div 15 = 300$.

Final Answer: 300 shillings

42. 4179450 seedlings in 1275 equal rows. Seedlings per row?

Step 1: $4179 \div 1275$: write 3; $3 \times 1275 = 3825$; $4179 - 3825 = 354$.

Step 2: $3544 \div 1275$: write 2; $2 \times 1275 = 2550$; $3544 - 2550 = 994$.

Step 3: $9945 \div 1275$: write 7; $7 \times 1275 = 8925$; $9945 - 8925 = 1020$.

Step 4: $10200 \div 1275$: write 8; $8 \times 1275 = 10200$; $10200 - 10200 = 0$.

Step 5: $1275 \times 3278 = 4179450$.

Step 6: $4179450 \div 1275 = 3278$.

Final Answer: 3278 seedlings

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Source Checks

Msomi Bora - Standard Five Exams | 2010 - 2023 | Monthly, Midterm, Terminal and Annual Exams - Download All Subjects - Msomi Bora